

Shikaiwen Sun

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1 Summary

Composer, pianist and researcher at the intersection of AI-generated music, complex rhythmic structures, and Music Information Retrieval (MIR).
Lead Composer of a human–AI co-creation project with collaborators from Music X Lab and G42. Experienced in reproducible research, model evaluation, and codebase replication.
Research interests: rhythmic complexity modeling, transformer-based music generation, human-centered evaluation of AI music.

2 Education & Academic Background

Queen Mary University of London (QMUL), 2025
MSc in sound and mussic computing (offer received)
Awarded the **Global Talent Scholarship**. Offer **deferred** due to preparation for direct PhD applications in Music AI / MIR.

Nanjing University of the Arts, 2018–2022
Bachelor of Arts (B.A.) in Composition and Composition Technology Theory (Computer Composition)
Undergraduate Thesis: Analytical Study of Kapustin's Eight Concert Etudes (focus on structure, technique, and stylistic features).
Masterclass Participation: Guest member of Prof. Gus Xia's masterclass at Nanjing University of the Arts in May–June 2023 and June–July 2024, collaborating with PhD students from NYU and MBZUAI.
Jazz Research Group Participation: Enrolled in Prof. Simon Dixon's Jazz Research Group at Queen Mary University of London from September 2025.

Mentor Backgrounds:
Prof. Gus Xia (Mohamed bin Zayed University of Artificial Intelligence, MBZUAI)
Associate Professor of Machine Learning. PhD from Carnegie Mellon University. Research interests include Machine Learning for Computer Music, Human-Computer Interaction, intelligent music systems, and interactive music composition agents.

Music and Computational Training

Music Coursework: Digital Audio, Recording, Music Production, Harmony, Polyphony, Analysis of Musical Forms, Orchestration (Western and Chinese National Orchestra).

Compositional Techniques:

- **Jazz and Contemporary Idioms:** Jazz improvisation, modal and blues scales, syncopation, polymeter, and layered rhythmic writing. (*Example:* Pat Metheny's *The First Circle*)
- **Contemporary Composition:** Aleatoric music — John Cage's *Music of Changes*; Serial and sequence-based structures — Ligeti's *Musica ricercata*; Sound-block and pointillism textures — *Lux Aeterna*.

- **Electronic and Spectral Composition:** Grisey's *Partiels*; timbre-based form and spectral thinking.
- **AI and MIR-based Composition:** Human–AI collaborative creation integrating MIR features (rhythm, scale, meter) and generative AI systems.
- **Undergraduate Thesis:** *Analytical Study of Kapustin's Eight Concert Etudes (Op. 40)* — fusion of classical form and jazz idioms.

3 Awards and Honors

- Outstanding Student Scholarship, 2018–2019, Nanjing University of the Arts Annual Student Scholarship
- Queen Mary University of London Global Talent Scholarship, 2025

4 Research Experience

- **Dreamscape at the Louvre Abu Dhabi (Lead Composer):** Lead composer for the world's first AI fully generated music score, in collaboration with AI scientists. Supported by G42 and MBZUAI. Executive Creative Director: Dr. Gus Xia. US pianist Dr. Gabriel Shao will premiere the work via human–AI interactive performance with a self-playing piano. Result: Demonstration of human–AI co-creation workflow. Portfolio: https://mxladicc25.github.io/.
- **CMPD (Classical Music + Reviews) Dual-Modal Dataset (ongoing):** Co-developing a dataset pairing classical works with critical reviews for machine learning research.

5 Professional & Technical Skills

- **Music:** Composition, Harmony, Counterpoint, Piano Performance, Orchestration
- **Technical:** Audio Signal Processing, Digital Audio, Recording
- **Programming:** Python (Librosa, Madmom, NumPy, PS2M), MATLAB
- **Research:** Reproducible environments, code replication, transformer/VAE architectures

6 Publications / Writing (Selected)

- CMPD dataset report (in preparation): dual-modal alignment and evaluation strategies.
- Undergraduate thesis: "Analysis of Kapustin's Eight Concert Etudes."